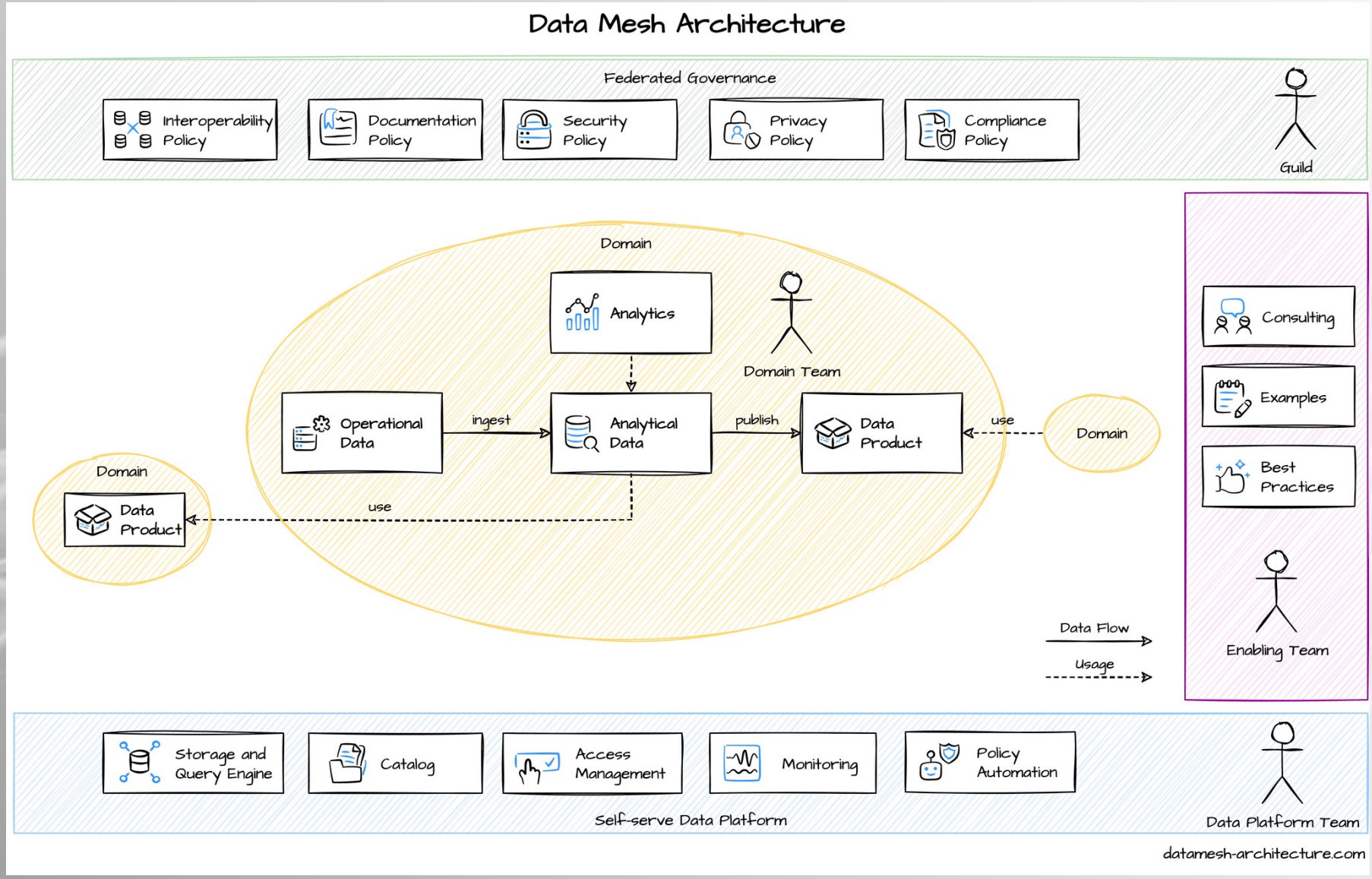
Data Lake vs Data Warehouse

GWDG

DATA WAREHOUSE	vs.	DATA LAKE
structured, processed	DATA	structured / semi-structured / unstructured, raw
schema-on-write	PROCESSING	schema-on-read
expensive for large data volumes	STORAGE	designed for low-cost storage
less agile, fixed configuration	AGILITY	highly agile, configure and reconfigure as needed
mature	SECURITY	maturing
business professionals	USERS	data scientists et. al.

https://www.kdnuggets.com/2015/09/data-lake-vs-data-warehouse-key-differences.html

Data Mesh by Dehghani



Data Fabric

- an **architectural concept**
- focuses on automating data integration, data engineering and governance in a data value chain
- is based on the concept of "active metadata”
- uses knowledge graphs, semantics and AI/ML technology to identify patterns in different types of metadata
- use these insights to automate and orchestrate the data value chain
- **complements a data mesh** and is not mutually exclusive
- improves the data mesh by creating data products faster, enforcing global governance and making it easier to control the combination of multiple data products.

Metadata

- is a must
- connects data with data or services
- necessary for control and efficiency when processing data
- required for discovery and access

Problem

- meta data is distributed in different places
- no standard format is used

Meta Data

- With given meta data the described data can be interoperable

- FAIR principles
  - Findable
  - Accessable
  - Interoperable
  - Reusable

- The principles emphasise machine-actionability … <> … because humans increasingly rely on computational support to deal with data as a result of the increase in volume, complexity, and creation speed of data.

https://www.go-fair.org/fair-principles/

META DATA

Consequences – FAIR

Findable

F1. (meta)data are assigned a globally **unique and eternally persistent identifier**

F2. data are described with **rich metadata**

F3. (meta)data are registered or indexed in a **searchable resource**

F4. metadata **specify** the data identifier

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Accessible

A1. (meta)data are **retrievable by their identifier** using a **standardized communications protocol**

A1.1. the **protocol** is open, free, and universally implementable

A1.2. the **protocol** allows for an authentication and authorization procedure, where necessary

A2. **metadata are accessible**, even when the data are no longer available

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Interoperable

I1. (meta)data use a **formal**, accessible, shared, and **broadly applicable language** for knowledge representation

I2. (meta)data use **vocabularies** that follow FAIR principles

I3. (meta)data include **qualified references** to other (meta)data

https://www.force11.org/group/fairgroup/fairprinciples

Consequences – FAIR

Re-usable

R1. meta(data) have a plurality of **accurate and relevant attributes**

R1.1. (meta)data are released with a clear and accessible **data usage license**

R1.2. (meta)data are associated with their **provenance**

R1.3. (meta)data **meet** domain-relevant **community standards**

https://www.force11.org/group/fairgroup/fairprinciples

Meaning of words

- Controlled Vocabular
  - a carefully selected list of words and phrases, which are used to tag units of information
- Taxonomies
  - a controlled vocabulary with a hierarchical structure
- Thesauri
  - terms have relationships to each other.
- Ontologies
  - Large thesaurus

Controlled Vocabular

- predefined and authorized terms
- preselected
- reduce ambiguities of natural language
- simple lists or complex taxonomies / thesauri
  - if relation between terms exist

- Can be used to
  - enable consistent indexing
  - improve the retrievability of documents
  - help to make searches more precise
  - serve to understand the contents of a domain
  - support interoperability

Taxonomies

Taxonomies are based on the general principles of classification:

- Structure of a subject area
- hierarchy formation
- Subordination of the specific to the general
- Ranking of subordinate terms

Thesaurus

Thesauri

- support consistent indexing
- facilitate the retrieval of index terms or search terms
- enable an expansion of the search
- store knowledge about the subject area
- serve interoperability

Types of Vocabularies

GWDG

List of Words	Taxonomy	Thesaurus	Ontology
• Control of the terms	• Control of the terms • Hierarchical relations	• Control of the terms • Hierarchical relations • Equivalence relation • Association relation	

Linked Data

By Tim Burners Lee

- Uniform Resource Identifiers (URIs) should be used to name and identify individual things.
- HTTP URIs should be used to allow these things to be looked up, interpreted, and subsequently "dereferenced".
- Useful information about what a name identifies should be provided through open standards such as RDF, SPARQL, etc.
- When publishing data on the Web, other things should be referred to using their HTTP URI-based names.

Linked Data II

Also T. B. Lee

- All conceptual things should have a name starting with HTTP.
- Looking up an HTTP name should return useful data about the thing in question in a standard format.
- Anything else that that same thing has a relationship with through its data should also be given a name beginning with HTTP.

Signposting

- "… is an approach to make the scholarly web more friendly to machines" signposting.org
- Can be used in HTTP Headers
- List of defined relation types:
  - https://www.iana.org/assignments/link-relations/link-relations.xhtml

```
HTTP/1.1 302 Found
Server: Apache-Coyote/1.1
Vary: Accept
Location: http://www.dlib.org/dlib/november15/vandesompel/11vandesompel.html
Link: <http://orcid.org/0000-0002-0715-6126> ; rel="author",
      <http://orcid.org/0000-0003-3749-8116> ; rel="author"
Expires: Tue, 31 May 2016 17:18:50 GMT
Content-Type: text/html;charset=utf-8
Content-Length: 217
Date: Tue, 31 May 2016 16:38:15 GMT
Connection: keep-alive
```

PERSISTENT IDENTIFIER

Persistent Identifiers

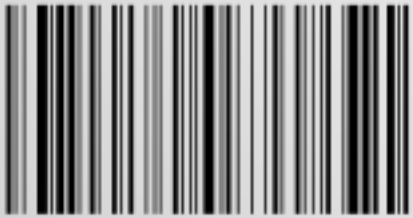
A PID is a

- standard, invariant and long-term reference
- of a digital resource
- regardless of status location or current owner

- Identifiers are commonly, everywhere used
- Passports
- Bank account numbers
- ...

- Which ID Systems do you know?

Examples



- Bar-Codes
- QR-Codes
- ISBN
- DOI https://www.doi.org
- PID http://www.pidconsortium.eu
- ARK https://confluence.ucop.edu/display/Curation/ARK
- Purl https://archive.org/services/purl/

Identifier Systems

- ARK
- arXiv
- PURL
- BIBCODE
- EAN-13
- eISSN
- IGSN
- ISSN
- ISTC
- IISSN
- LSID
- PMID
- UPC
- Handle
- DOI
- URN
- URL
- URI
- RoR
- RAID
- ORCID
- ……

Actionable IDs

From:

A Persistent Identifier (PID) policy for the European Open Science Cloud

(doi: 10.2777/926037)

Machine Actionable:

- Machine Actionable means that a formal statement is syntactically and semantically specified enabling computing systems to carry out automatic processing.

URLs

- We use URLs e.g. www.google.com in our daily life
- URL: Uniform Resource Locater are a subset of URIs (Uniform Resource Identifier)

https://max:muster@www.example.com:8080/index.html?p1=A&p2=B#ressource

- Defined by RFC 3986 https://tools.ietf.org/html/rfc3986

URLs and DNS

- URLs have to be translated to IP adresse in order to find the server
- Domain Name System translates to IPs

www.google.com

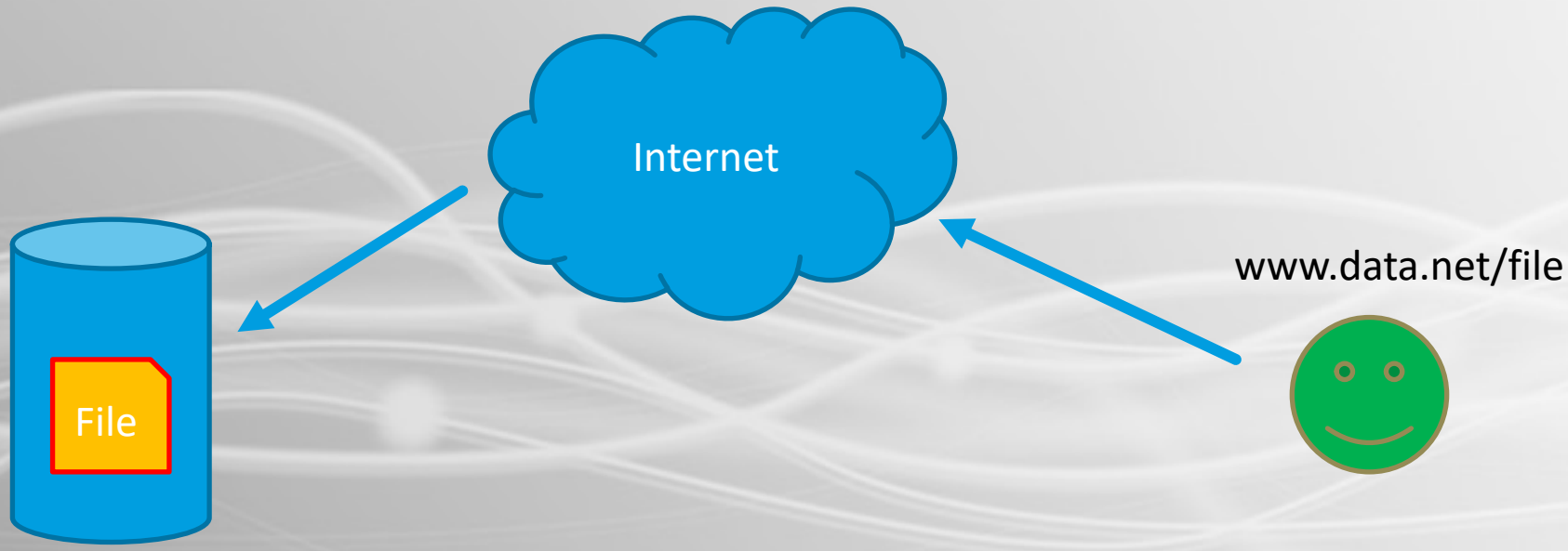
- Or from the slides:

www.intel.de/content/www/de/de/big-data/big-data-101-animation.html

Problem: URLs often change!!!

Problem Statement

- Data is referenced with an URL



Problem Statement

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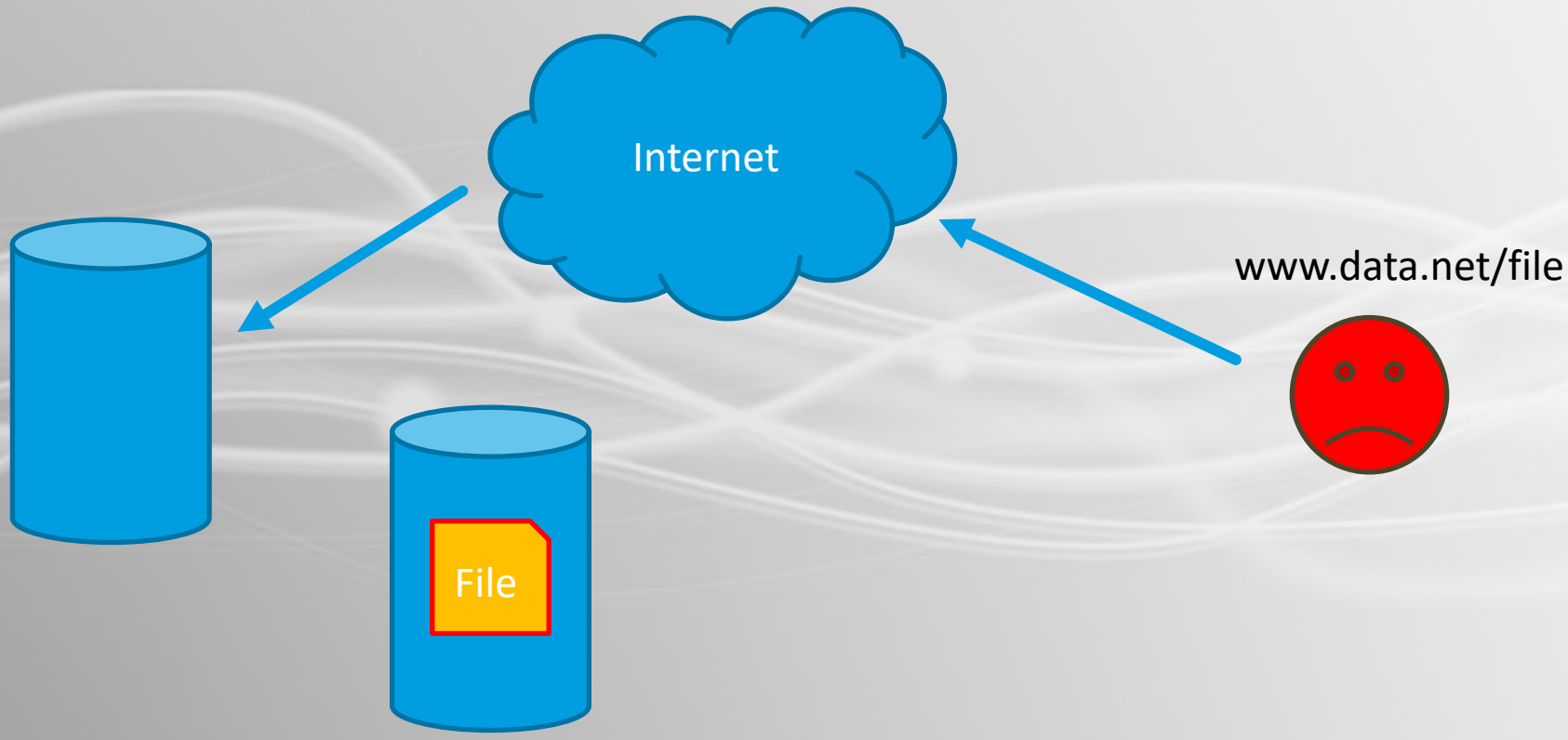
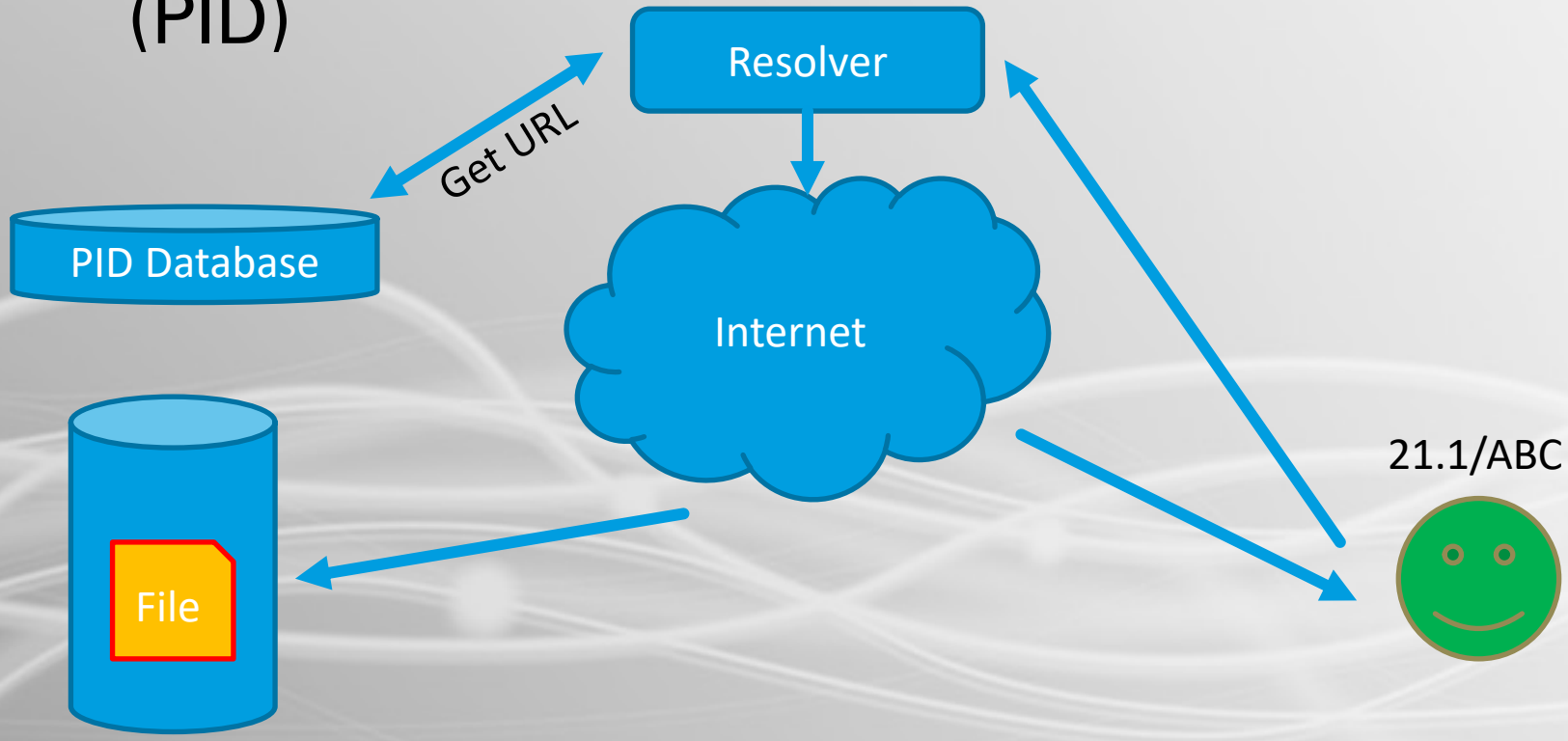


Problem Statement

- Instead of URL use a Persistent Identifier (PID)



Problem Statement

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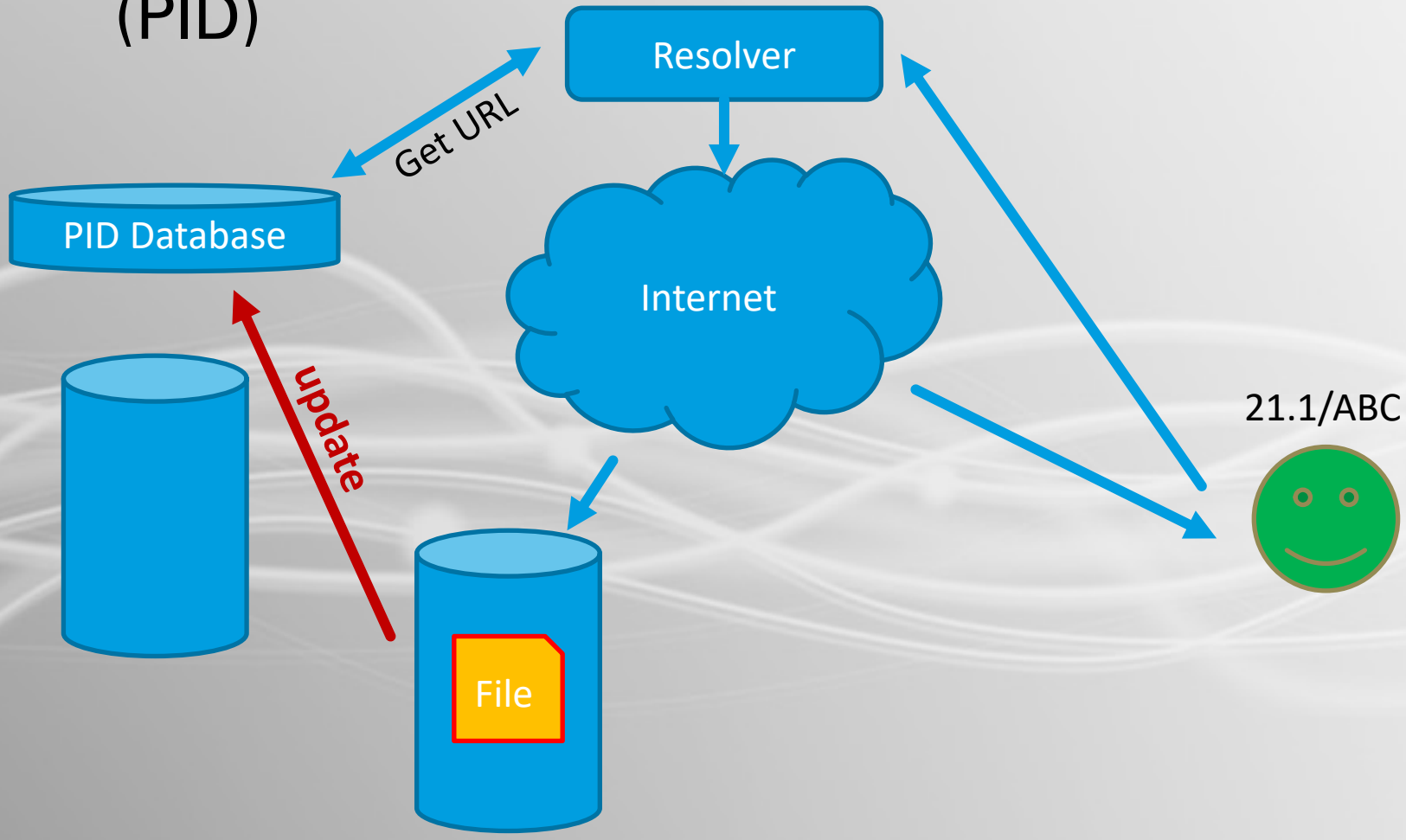
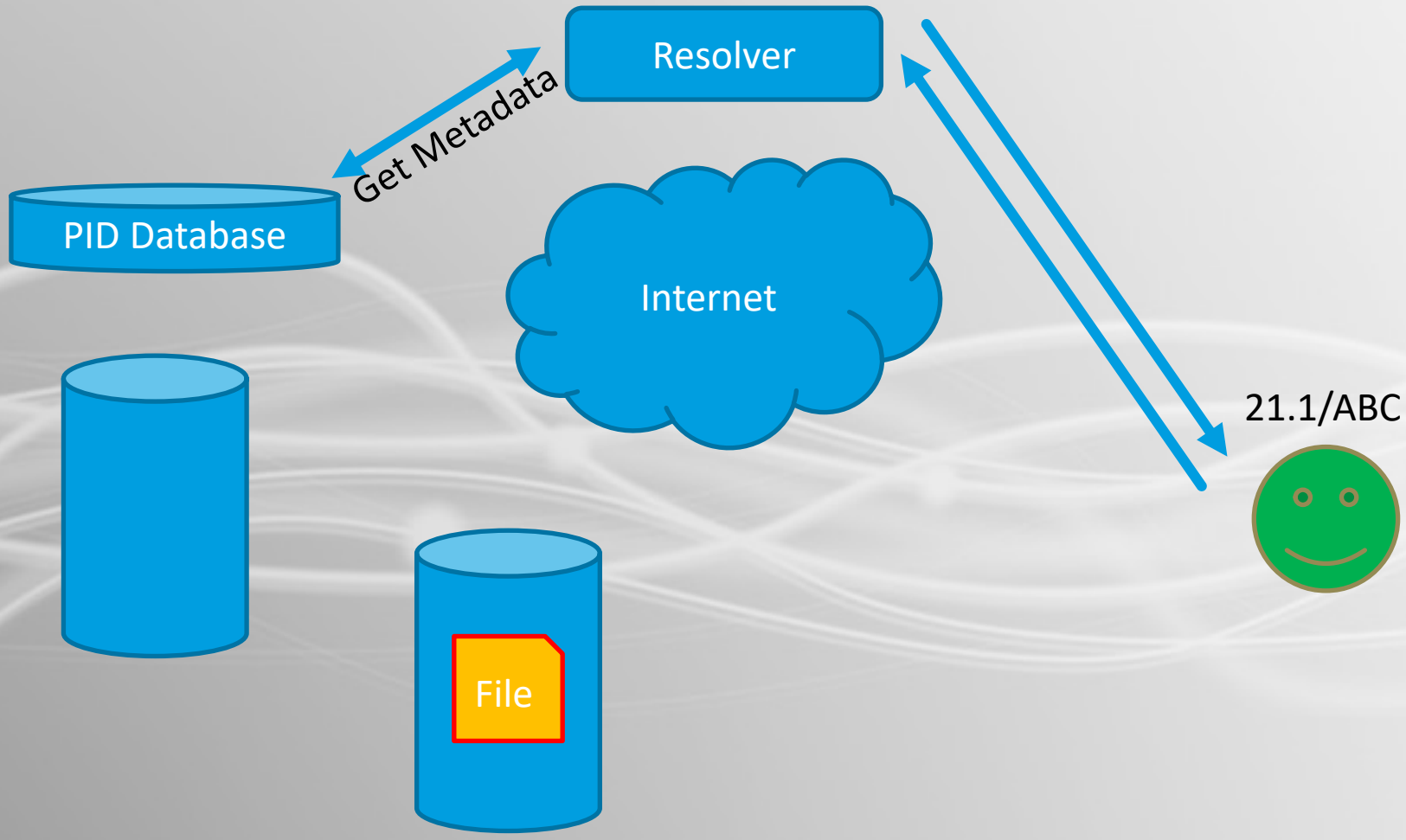


Problem Statement

- Learn about the data without download



Persitent Identifier

- We need a stable reference to a resource
- The reference does not need to be human readable
- A Persistent Identifier points to a resource

PID:

21.11101/1234567

DOI Example

Astronomy & Astrophysics

Worldwide astronomical and astrophysical research

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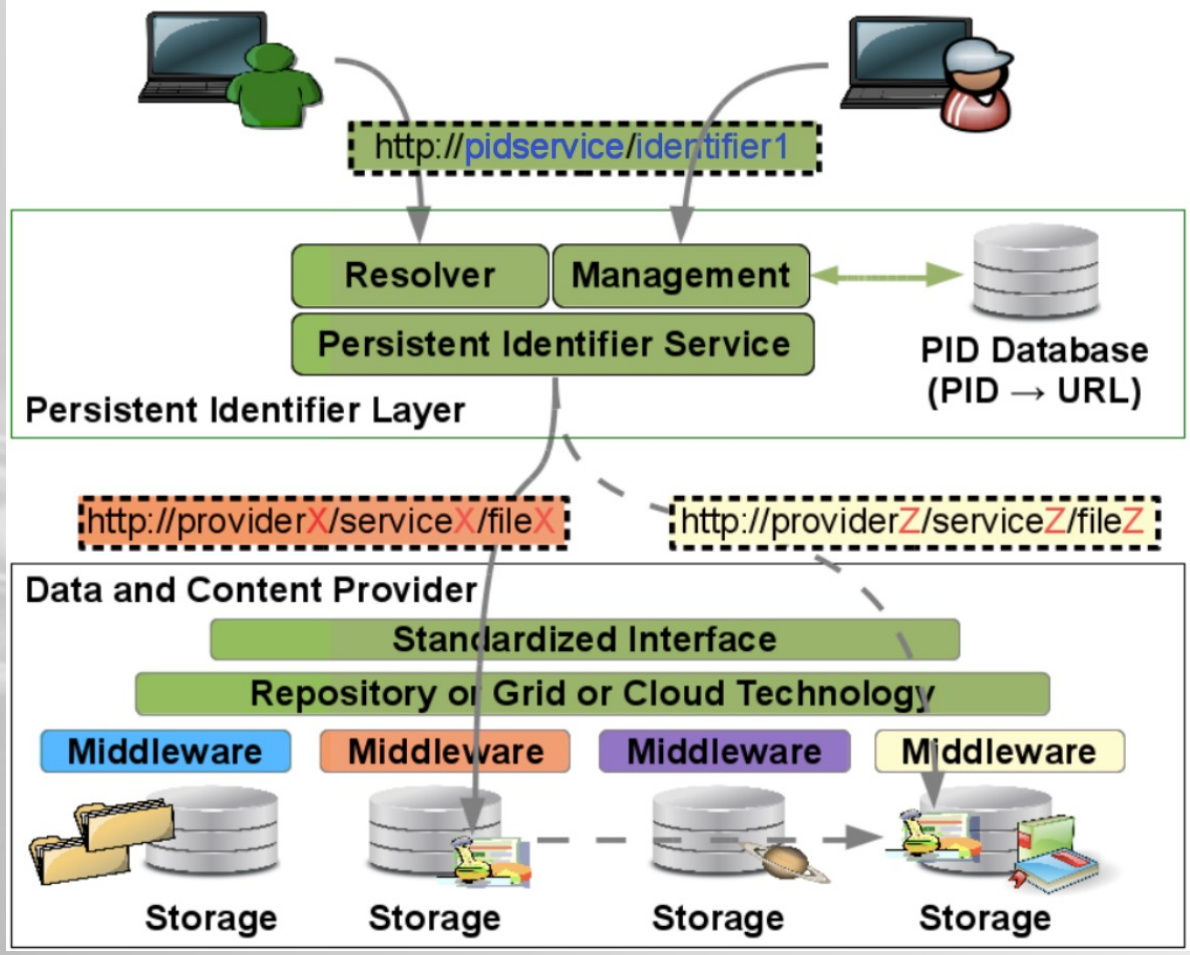
Free Access

Issue	A&A **Volume** 530, June 2011
Article Number	A112
Number of page(s)	12
Section	The Sun
DOI	https://doi.org/10.1051/0004-6361/201016019
Published online	20 May 2011

Use Case

- Defined interface between reference and object
- The reference is resolved by the PID Service, which gives the address of the PIDs objects actual position
- If the object does not exist anymore, this also can be given inside the PID
- Additional of additional Services are possible



Sustainability by technology

- Management and resolving of PIDs must be guaranteed for a long period of time
- The PID System must be
  - reliable
  - robust
  - and operational on long term perspective
- Many organisation use well tested systems:
  - Handle, DOI, URN, ARK ...
  - Example: PIDs of ePIC

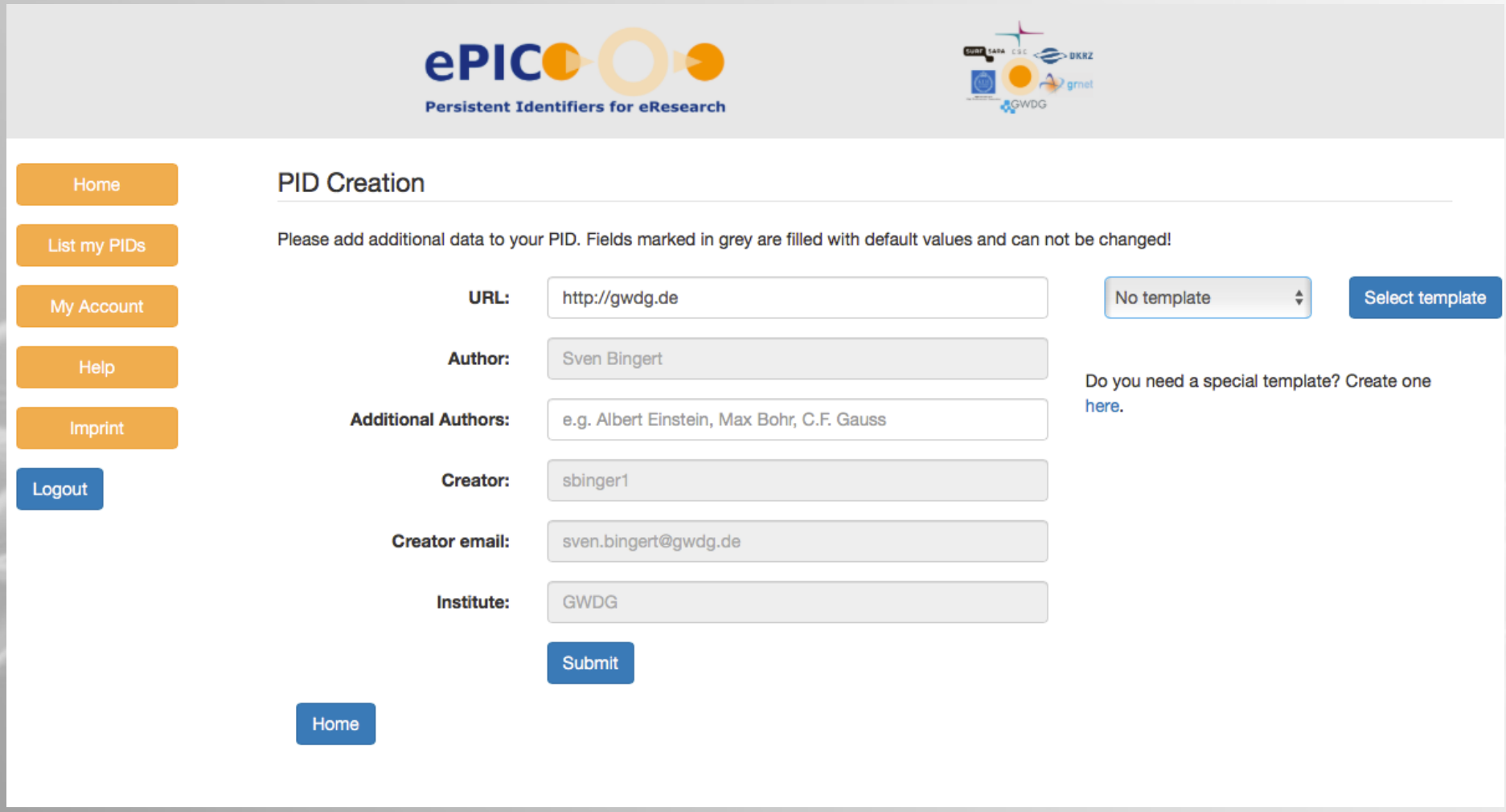
Handle System ©

- World wide system to create, update and resolve PIDs
- It consists of two parts:
  - the registration of prefixes and
  - the software
- HANDLE.NET Software v9
- Managed by Digital Object Numbering Authority (DONA)
  - Non-profit Organisation for the administration of the handle system
  - Multi-Primary Administrator (MPA)
    - currently 9
    - each responsible for a specific name space

ePIC API V2

- To enforce specific policies (e.g. no-delete)
- REST API
- Documentation
  - http://doc.pidconsortium.eu
- Free choice of suffix structure
- Distributed system of primaries and mirrors for high availability service

ePIC Webformular



ePIC PID vs. DOI

- ePIC PID and DOI are based on handles.
- Both, ePIC PID and DOI can be resolved globally without knowledge of the home institution.
- Neither ePIC PID nor DOI can be deleted.
  - Handles can be deleted
  - matter of policies of ePIC and DOI
- *Difference*: DOI
  - redirects to landing page
  - Meta data in a separate data base